WMA CTO

Setting 2011 Agenda and Priorities

8 January 2010

REVIEW DRAFT 2011-01-08A

Context
Business Demand and WMA CTO Mission/Commitments

- Key priorities for 2011
  - Supporting strategic products
  - Enhancing FA productivity
  - Providing a compelling client experience
- TSA + Subsequent workshops provided clear set of “TODOs” for WMA CTO
- WMA CTO also contributes to Group CTO activities [to (among others) out-smart and out-standardize the competition] eg,
  - Enterprise Target Architecture
  - Leverage and share technology and infrastructure to accelerate change
  - CTO Workgroups including ETC etc…
- We are also used as “Connecting the dots,” an effort we will not attempt to quantify!
- Key governance processes must be defined/managed to
  - Track progress against the target state
  - Assure the alignment of technology investments with the strategy and business priorities
- WMA CTO will support change management
  - Fostering efficient development processes and cultivate innovation
  - Instill a disciplined approach to architecture and system design in order to significantly advance the target state objectives in 2011

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Process Discussions

As discussed at the Tech Council developing and using the right processes is paramount…

Questions

- **Do we all agree that the WMA CTO Function Cover These Processes?**
- **What Are the Priorities?**

Implementation

- **Owning Process =**
  - Developing the Playbook and other Reference Documentation
  - Running the process
  - Create reusable Artifacts
- **Approach**: Develop Playbook iteratively, on the back of one initiative

Legend

- (O = Own / S = Shared)
- (*) = Cross UBS Linkage

- **Maintaining and Promoting TSA** (O) (*)
- **Monitoring implementation of TSA** (O) – ie "Air Traffic Control" function with Linkage with Development, CFO etc… teams
- **Investment Review / Governance** (O)**:**
  - Initial Funding/TIC Stage
  - Architecture Review throughout execution of Projects
- **ETC / Category Management** (O) (*)
  - Within WMA
  - Representing WMA at UBS-level ETC
- **Reference Architectures Development** and **Implementation Governance**, **Syndication** (O) (*)**:**
  - Data Architecture
  - Data Services framework
  - Services Architecture
  - Use Interface Architecture
  - Security and Entitlements Framework
  - Service Management (Requirement gathering)
- **Technology Evaluation**
  - **Ad hoc research/evaluation of technologies** (pre-POC) (Example: Endeca) (O/S) (*)**:**
  - **Management of POCs** (O/S) (*)**:**
  - **Suitability Analysis to import technology from/to WMA**(eg F35) (O) (*)**:**
- **Complex Offering Product Management** (O) (*) to frame what we want to do (Candidate: BPM/Workflow)
- **Internal Communication** (O) - Reach Out Program to Technologists (eg HTML 5 Col
- **Actual Project Management** (O)
  - Major TSA and non-TSA-related Initiatives (See Attachment A)
  - gCTO Work Groups (O) (*)
  - Support WMA-Specific Major Initiative (eg Rent an Architect/Strategist)
- **Innovation and Other Intellectual Capital Development** (O) (*)
- **Strategic Cost Optimization (vs On-Going Cost and Demand Management**
- **SDLC Definition and Joint Support**

REVIEW DRAFT 2011-01-08A

… but from a practical standpoint, this is not just about Process Creation and Management – Major Initiatives are underway

TSA Related (Ranked by Strategic Importance) See Attachment A (and C)

02-02 - All Client Data read from CRDB
02-03 - All Financial Instrument Data read from PRDB
02-04 - All Client Transaction Asset & Liability Data read from CPOS
02-05 - All Transaction Data read from OET
01-08 – Set Up Data Governance
02-01 - Convert to Universal Account #
01-06 - Client Transaction, Asset & Liabilities and Transaction Master
01-04 - Transaction Master
01-05 - Entitlements Master
04-04 - Enhanced Account Opening Process
01-03 - Client Master
04-01 - Rewrite Client Frontends
04-02 - Rewrite FA Frontends
04-03 - Source all Client Reports from Gold Sources
08-02 - Enhance Annuities & Insurance Platform
01-01 - Establish Data Service Framework
09-01 - OneTransactionKey
06-01 - One Client Profile
04-05 - Client Entitlement and Role-Base services
03-03 - Replace Core Platform
07-03 - Upgrade Electronic Billpay
07-01 - New Cards Platform
07-02 - Enhance Mortgage Platform
08-04 - Stock Loan
08-05 - Structured Products
08-06 - CEFS
03-04 - Front to Back Workflow Engine
05-01 - Consolidate Order Entry Applications
01-09 - Virtual Data Warehouse
01-02 - Financial Instrument Master
06-02 - Integrated Wealth Management Platform
04-08 - Integrated Contact Management Application
08-01 - Alternative Investment Automation
04-07 - Usability, Touch and Feel
05-03 - IB Integration
01-07 - Client Document Master
08-03 - Mutual Fund Back office
04-10 - Computer Telephony Integration
03-05 - Convert to Group Reconciliation Utility

Others Legacy / Carry-Over and By-Product of Post TSA Discussions and Strategic Workshops (Need to define Own vs Support)

- **WMA Specifics**
  - SDLC /Agile Review
  - Client Facing Technology
  - Project JUNO
  - Enterprise Notification Services (Pub/Sub)
  - (Real Time) Risk Analytics
  - Workflow
  - CEFS Global
  - Cost-cutting initiative 2011-2012
  - Wealth Management Framework
  - Security and Risk – assess CRDB and MAC
- **Participation in CTO / UBS-wide Work Groups and Cross UBS Exchange Options**
  - Data Design Authority
  - MDS Optimization (Market Data)
  - Cost Take-Out
  - Client Facing Technology
  - Real-time Transaction Processing- Sharing assessment
  - F35 architecture assessment
  - Mobile Strategy
- **POCs**
  - macBook Air
  - Cicero
  - Mobile apps
  - HTML5
  - Netezza POC
  - Composite for GMIS
  - Client 360
- **Others: CTO Strategy/Marketing/Com function development**
  - Line-up with WMA:
    - Product Development
    - Innovation Playbook
    - Communication Function development
  - Others

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Activities Deliverables (Process)

Process		Deliverables		
		End Month (ie for Meeting on 25th)	Q1	2011
Maintaining and Promoting TSA (Conceptual)		· Brief EDs + DIs (TSA) · Plan TSA Com. Plan (Attachment B) + Town Meeting on 23rd · Explain CTO-led Governance Model to 1/25 Participants	· Execution of Com Plan · Broader Communication on TSA and Governance model	Release 2.0 (Mid-Year)
Monitoring implementation of TSA		Review and get agreement on "Air Traffic Controller" execution model	· Implement ATC incubated in WMA CTO · Define and Approve KPIs · Publish Baseline Dashboard	Process moved from CTO to COO
Architectural Review	Initial Funding/TIC Stage	· Develop guidance 0.0 for TIC Review (License too Build) · Review TIC Process · Identify "Clear and Present Danger" Items, · Identify buckets of decision types (Mandated vs Recommended etc…)	· Publish 1.0 Guidance for TIC · Updated Process (Longer Review Process) · Estimati9on Tool Methodology · + Support set up of QA function · Negotiate Fundings	· Process Implemented · Estimation methodology in place · Incorporate Guidance in 2012 Investment Process
	Throughout execution of Projects	SDLC Review Dependant		
ETC and Category Management	Within WMA	Plan development of: · SW footprint analysis (including upcoming renewals + EOL) · Product/Solution Roadmap	· SW footprint analysis (including upcoming renewals + EOL) · Product/Solution Roadmap framework · GTIS Implications framework (Shared deliverable)	· Product/Solution Roadmap framework · Joint capacity plans with GTIS
	Representi WMA a (UBS-level)	On-Going (see above WMA contributions)		
Architectural Reference Architecture and Framework Development / Management	Data Architecture and Data Services framework	· Identify and Prioritize elements of the Data Journey · FI Data Rationalization cost estimate	· Data Governance Model Defined and Implemented · Enterprise Data Model –Funding asnd Execution Plan developed	Meta-Data definition (Assuming Funding)
	Services Architecture		· Reference Architecture Published (1.0) · Target Implementation Plan Developed with RoIT	Pursue Implementation
	Use Interface Architecture	· Kick off usability study and link with CFT initiative · CFT WMA Contribution for UK Presentation	Plan impact on framework development	Implement
	Service Management Architecture	· Current Status quick survey	Requirement gathering	Imput into SM process and current design/architcture
	Security and Entitlements Framework	TBD		
Technology Evaluation	· Ad hoc research/evaluation of technologies (pre-POC) (Example: Endeca) · Suitability Analysis to import technology from/to WMA (eg F35)	Demand forecast and staffing estimates	Publish Process	
	Management of POCs (O/S) (*)	Selection of POCs	Publish Playbook	
Complex Offering Product Management		Concept Presentation	Pilot Concept on BPM /Workflow in WMA	Extend to other complex offerings
Internal Communication / Reach Out Program to Technologists			· Develop Plan · First Townhalls	Continue Execution
Innovation and Other Intellectual Capital			· Innovation Playbook	

REVIEW DRAFT 2011-01-08A

In addition, pending a comprehensive cross- WMA IT planning exercise (1/25/2011), we developed a short list of deliverables for Q1 that depend on funding and others that are currently underway

From a Practical Standpoint this is not just about Process and Great Management – We have real Major Initiatives to support and Plan

TSA Related (Ranked by Strategic Importance) See Attachment A

0202- All Client Data read from CRDB
0203- All Financial Instrument Data read from PRDB
0204- All Client Transaction Asset & Liability Data read from CPOS
0205- All Transaction Data read from OET
0108- Set Up Data Governance
0201- Convert to Universal Account #
0106- Client Transaction, Asset & Liabilities and Transaction Master
0104- Transaction Master
0105- Entitlements Master
0404- Enhanced Account Opening Process
0103- Client Master
0401- Rewrite Client Frontends
0402- Rewrite FA Frontends
0403- Source all Client Reports from Gold Sources
0802- Enhance Annuities & Insurance Platform
0101- Establish Data Service Framework
0901- OneTransactionKey
0601- One Client Profile
0405- Client Entitlement and Role-Base services
0303- Replace Core Platform
0703- Upgrade Electronic Billpay
0701- New Cards Platform
0702- Enhance Mortgage Platform
0804- Stock Loan
0805- Structured Products
0806- CEFS
0304- Front to Back Workflow Engine
0501- Consolidate Order Entry Applications
0109- Virtual Data Warehouse
0102- Financial Instrument Master
0602- Integrated Wealth Management Platform
0408- Integrated Contact Management Application
0801- Alternative Investment Automation
0407- Usability, Touch and Feel
0503- IB Integration
0107- Client Document Master
0803- Mutual Fund Back office
0410- Computer Telephony Integration
0305- Convert to Group Reconciliation Utility

Others Legacy / Carry-Over and by Product of Post TSA Discussions and Strategic Workshops Needs to define Own vs Support

- **WMA Specifics**
  - SDLC /Agile Review
  - Client Facing Technology
  - Project JUNO
  - Enterprise Notification Services (Pub/Sub)
  - (Real Time) Risk Analytics
  - Workflow
  - CEFS Global
  - Cost-cutting initiative 201-2012
  - Wealth Management Framework
  - Security and Risk- assess CRDB and MAC
- **Participation in CTO / UBSwide Work Groups and Cross UBS Exchange Options**
  - Data Design Authority
  - MDS Optimization (Market Data)
  - Cost TakeOut
  - Client Facing Technology
  - Real-time Transaction Processing Sharing assessment
  - F35 architecture assessment
  - Mobile Strategy
- **POCs**
  - macBook Air
  - Cicero
  - Mobile apps
  - HTML5
  - Netezza POC
  - Composite for GMIS
  - Client 360
- **Others: CTO Strategy/Marketing/Com function development**
  - Line-up with WMA:

A High Level (Year/Year) Timeline was Developed (See Attachment C).

Here are "No Regrets Items" for Q1

Other Activities	Q1	2011
SDLC /Agile Review	Identify Partner (O)	TBD
Client Facing Technology (WMA)	Reference Stack (O)	Implementation RoadMap ("IR") (S)
Project JUNO	Evaluation (O)	IR (S)
Enterprise Notification Services (Pub/Sub)	Ref. Architecture (O)	Product Definition (O) Tech. Implementation (S)
(Real Time) Risk Analytics	TBD	
Workflow	Reference Arch. (O) Product Definition (?) (O)	IR (S)
CEFS Global	Syndicate Findings (O)	IR (S)
Cost-Cutting initiative 2011-2012	TBD	TBD
Wealth Management Framework	Business Architecture (O)	TBD
Security and Risk – assess CRDB and MAC	Initiate	TBD
Data Design Authority	Workgroup Contribution (Surveys etc…) = WGC	
MDS Optimization (Market Data)	WGC	
Real-time Transaction Processing	Sharing Assessment	TBD
F35 architecture	Sharing Assessment	Pilot (TBD)
Mobile Strategy	Pilot Preparation	Pilot
macBook Air	Beta	Pilot / TBD
Cicero	POC	TBD
HTML5	CoI Support	TBD
Netezza	POC	TBD
Composite	GMIS + / Client 360 Pilots	TBD
Product Development	Function Descriptin + Pilot	Deployment
Innovation Playbook	Discussion Document	Implementation / TBD

	Q1 Spend ($m)	Est. Jan Spend	HC	Notes
IT Foundational Projects				
Extension of Wipro Data Initiatives	$0.3	$0.3	0	2 Extensions for 297k; Important to Approve by Dec 2
Advisor Desktop Usability Assessment	$0.5	$0.4	2	Estimate
Data Rationalization work	$0.5	$0.0	7	Assumes 50% vendor
Build Data Services	$0.5	$0.0	7	Assumes 50% vendor
Cognos Migration	$0.5	$0.2	7	Assumes 50% vendor
Trading Platform Analysis / Assessment	$0.5	$0.3	3	Assumes 75% vendor
WMI Canada / WMI Application Assessment	$0.5	$0.2	3	Assumes 75% vendor; Trying to get Canada portion fo

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Delivery Model and Options

The ability to deliver on the Group CTO and Stream CIO priorities will be driven by the team's capacity. Three scenarios are laid out here:

	Scope of Activities	Scenario A: 8 FTEs impact	Scenario B: 10 FTEs impact	Scenario C: 12 FTEs impact	Ideal Skill Set and FTE Total	Mgment	TA	AA	PM	*US*	
Group CTO	Support Group CTO Quality Process Reviews				1.1	0.2	0	0	0	*1.1*	+ 3 to 6 FTE (External Help)
	Promote Group Tech Arch; Category Mgt				0.3	0.1	0	0	0	*0.3*	
	Achieve Group Tech Council Objectives				2.1	0.8	0	1	0	*2.1*	
	Total				3.5	1.1	1	1	1	*3.5*	
WMA CTO	Promote Innovation and R&D				0.7	0.1	0	0	0	*0.7*	
	Drive WMA Target State Architecture				4.5	0.8	1	1	2	*4.5*	
	Total				5.2	0.9	1	1	2	*5.2*	
Foundational - WMA CTO	Support Vibrant IT Community				0.5	0	0	0	1	*0.5*	
	Support Development of UBS-wide Architectures				0.2	0.1	0	0	0	*0.2*	
	R&D for new and shared technologies				2.2	0.1	1	1	1	*2.2*	
	Develop IT Knowledge and Competitive Advantage				0.6	0.1	0	0	0	*0.6*	
	Total				3.5	0.3	1	1	1	*3.5*	
	Total				12.2	2.3	3	4	4	*12*	

LEGEND

Risk to advancing Group CTO/ WMA IT Strategy

Concerns on ability to drive towards strategic goals

Will deliver on strategic goals

See attachment D for mapping of Activities in broad domains depicted in this tables and staffing assumption

REVIEW DRAFT 2011-01-08A

Short Term Objectives

In the 30 Days, our TODO List is Clear (In addition to other Book of Work Items)

1. Debrief all 1/25 participants and most EDs + DIs on TSA (with 45 days)
2. Explain CTO-led Governance Model to 1/25 Participants and most EDs + DIs on TSA (with 45 days)
3. Propose an extended TSA Communication Plan
4. Prepare a presentation on “Air Traffic Controller” execution model to be discussed at the meeting
5. Develop a check list for TIC Review (License to Build) based on gCTO model
6. Review current TIC Process and propose changes to be discussed on the 25th
7. Identify “Clear and Present Danger” Items and act on them (On-Going)
8. Identify buckets of funding decision types (Mandated vs Recommended etc…)
9. Plan the development of:
   - SW footprint analysis (including upcoming renewals + EOL)
   - Product/Solution Roadmap
10. Identify and Prioritize elements of the Data Journey
11. Get a FI Data Rationalization cost estimate
12. Agree on sourcing of usability study and linkage with CFT initiative
13. Finalize/rationalize the list of POCs we will pursue
14. Discuss the model for promoting Innovation
15. Agree on how CTO contributes to Security and Entitlement framework discussions and SDLC/QA process re-engineering

Need to Prioritize!

REVIEW DRAFT 2011-01-08A

Q1 = New Book of Work

Process - Related

1. Broader Communication on TSA and Technology Governance Model throughout WMA IT
2. Implement ATC incubated in WMA CTO
3. Define and Approve KPIs for TSA implementation tracking and Publish Baseline Dashboard
4. Architecture Review Process (pre-TIC) in-place
5. Estimation Methodology 1.0
6. QA function support
7. Funding and budget finalized
8. SDLC: TBD
9. SW footprint analysis (including upcoming renew als + EOL)
10. Product/Solution Roadmap framew ork
11. GTIS Implications framew ork (Shared delivera
12. Participation to gCTO governance process
13. Data Governance Model Defined and Implemented
14. Enterprise Data Model –Funding and Execution Plan developed
15. Service Architecture Reference Architecture Published and Target Implementation Plan Developed w ith RoIT
16. POC Playbook published
17. Pilot Product Management Concept on BPM /Workflow in WMA
18. Innovation Playbook
19. New Edition of “Innovation Program/Challenge” planned

Others

1. Identify a partner to promote SDLC /Agile Review
2. Publish Client Facing Technology Reference Stack
3. Evaluate usage of JUNO in WMA
4. Publish an Enterprise Notification Services (Pub/Sub) Ref. Architecture
5. Perform TBD w ork in (Real Time) Risk Analytics
6. Publish a Reference Architecture and Product Management Artifacts for Workflow /BPM
7. Syndicate finding for CEFS Global and secure next step decision
8. Contribute to Cost-Cutting initiative 2011-2012
9. Finalize a first version of the Wealth Management Framew ork Business Architecture
10. Assess CRDB and MAC linkage
11. Join the Data Design Authority and contribute to its activities
12. Contribute to the MDS Optimization (Market Data) w orkgroup activities
13. Assess Real-time Transaction Processing sharing options
14. Assess F35 architecture and sharing options
15. Prepare pilot for mobile service
16. Implement macBook Air Beta (MiniPOC)
17. Perform Client 360 bake off (Cicero, Composite, CW/Widgets-based, CW/Vignette (legacy), F35)
18. Perform Cicero POC and propose next steps (eg Stractical 360 option)
19. Officialize HTML5 CoI
20. Perform Netezza POC and propose next steps
21. Perform Composite GMIS + / Client 360 Pilots + analysis of Book Query
22. Product Management Function Description + Pilot Deployment
23. Innovation Playbook Discussion Document
24. Community Development 2.0 Function Description

REVIEW DRAFT 2011-01-06A

Attachments

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A – Current Investment Requests
Background / Summary

- References:
  - The TSA document included a series of 41 activities in 9 domains that the WMA CTO organization, along with the WMA IT leadership identified as critical to the implementation of the TSA
  - The Key Change Workshop activities sponsored by WMA COO identified and prioritized the execution of approximately 130+ investments ("Strategic Investment") aligned with WMA business objectives
- As a result of this analysis, all stakeholders agreed that activities aimed at accelerating the implementation of the TSA ("CTO Initiatives") should fall into two categories:
  - "Embedded/ Vertical" support – i.e. augmenting the individual work plan for the Strategic Investments so that their implementation not only align with the TSA but also accelerate progress toward the target state (Example: making sure that the workflow-oriented project for disbursement also includes provision for BAM features)
  - "Foundational/Horizontal" activities – Investments (including POCs, external consulting and HW/SW acquisitions) that are best managed as cross-initiatives (Example: implementing the "Data Journey", ie the coordinated convergence of data structure and access methods to conform with the TSA tenets)
- This release 0.0 presents a high-level view of what we propose to remark in both categories along with a initial mapping of these CTO Initiatives with the portfolio of Strategic Initiatives

REVIEW DRAFT 2011-01-08A

A – Current Investment Requests
Proposal Summary

- The 2011 proposed spent for the overall Embedded and Foundational investment is 11.5M – an amount that is commensurate with the overall order of magnitude of the total WMA Investment portfolio (app. 10%) and the estimated ability of the WMA IT and CTO organization to effect change through these investments
- It is expected that similar investment will be required over the next 2 years in 2012 and 2013
- Of the $11.5M, $3.75M should be incorporated in the proposed Strategic Investment spent in the form of high-impact focused activities in the key core system transformation projects as well as the Workflow (eg BAM, developing “Workflow as a Product” construct, overall front end usability analysis etc…)
- The remaining investment ($7.75) consists of "horizontal" work on the Data Journey, Presentation Layer, and Services Oriented Architecture foundations.
- In terms of immediate requests (no regrets), the follow-items should be considered in priority:
  - Continuation of Wipro engagement to complete documentation of enterprise data model and retirement of duplicative data stores
  - Proofs-of concept related to presentation layer and mobile application development
  - Proofs of concept related to data journey as outlined in the Target State Architecture (eg virtual integration and federated data warehouse)
  - Assessment of F35 for WMA
- The team is working towards the development of a more granular analysis with quarterly plans and details for investment which will be presented to the TIC, similarly as all other investments

REVIEW DRAFT 2011-01-08A

A – Current Investment Requests

TSA-Initiatives (Prioritized) and Budget Request

Activity	Score	Normalized	Built-In	Cross-Project	Comments
02-02 - All Client Data read from CRDB	**52.5**	**0.78**			
02-03 - All Financial Instrument Data read from PRDB	**52.5**	**0.78**			
02-04 - All Client Transaction Asset & Liability Data read from CP(	**52.5**	**0.78**			
02-05 - All Transaction Data read from OET	**52.5**	**0.78**	$2.00		Core battle field for data converg
01-08 - Data Governance	**36.5**	**0.55**		$0.10	
02-01 - Convert to Universal Account #	**35.8**	**0.53**	$0.25	$0.25	
01-06 - Client Transaction, Asset & Liabilities and Transaction Ma	**35.4**	**0.53**			Included above
01-04 - Transaction Master	**31.8**	**0.47**			Included above
01-05 - Entitlements Master	**26.4**	**0.39**		$0.50	
04-04 - Enhanced Account Opening Process	**25.8**	**0.39**			
01-03 - Client Master	**25.3**	**0.38**		$0.50	
04-01 - Rewrite Client Frontends	**23.8**	**0.36**		$0.50	
04-02 - Rewrite FA Frontends	**23.8**	**0.36**		$0.50	
04-03 - Source all Client Reports from Gold Sources	**23.8**	**0.36**		$0.50	
08-02 - Enhance Annuities & Insurance Platform	**23.4**	**0.35**			
01-01 - Establish Data Service Framework	**23.3**	**0.35**		$0.50	
09-01 - OneTransactionKey	**23.2**	**0.35**		$0.50	
06-01 - One Client Profile	**21.8**	**0.33**		$0.50	
04-05 - Client Entitlement and Role-Base services	**21.7**	**0.32**		$0.50	
03-03 - Replace Core Platform	**21.5**	**0.32**			Included Above
07-03 - Upgrade Electronic Billpay	**21.4**	**0.32**			Depends on Banking
07-01 - New Cards Platform	20.0	0.30			
07-02 - Enhance Mortgage Platform	20.0	0.30			
08-04 - Stock Loan	20.0	0.30			
08-05 - Structured Products	20.0	0.30			
08-06 - CEFS	20.0	0.30			
03-04 - Front to Back Workflow Engine	19.5	0.29	$0.50		R100; R101;R103;R104
05-01 - Consolidate Order Entry Applications	18.6	0.28			
01-09 - Virtual Data Warehouse	18.5	0.28		$0.50	
01-02 - Financial Instrument Master	18.4	0.27		$0.50	
06-02 - Integrated Wealth Management Platform	17.8	0.27			
04-08 - Integrated Contact Management Application	16.6	0.25			
08-01 - Alternative Investment Automation	16.5	0.25			
04-07 - Usability, Touch and Feel	16.4	0.24	$0.50		Re: Outcome of Workshop
05-03 - IB Integration	16.2	0.24		$0.50	Re: EQI
01-07 - Client Document Master	15.4	0.23	$0.50	$0.50	Re: Outcome of Workshop
08-03 - Mutual Fund Back office	15.4	0.23			
04-10 - Computer Telephony Integration	14.6	0.22		$0.50	R119
03-05 - Convert to Group Reconciliation Utility	13.3	0.20			
05-03 - New Managed Acct Trading Platform	12.8	0.19			
03-01 - Real-Time Positions and Balances	12.7	0.19		$0.40	
04-09 - Practice Management Tools	11.0	0.17			
01-10 - Improve Business Intelligence Platform	11.0	0.16			
09-02 - Common Error Handling	10.2	0.15			
04-06 - Improved Performance Reporting Aligned with Client Goals	10.2	0.15			
03-06 - Consolidation of Fee Billing Applications	8.0	0.12			
03-02 - Remove Margin from Bookkeeping	3.9	0.1			
Total:			$3.75	$7.75	

REVIEW DRAFT 2011-01-08A

Attachment B

TSA Elements of a Communication Strategy

- Who
  - Project Managers/Application Leads
  - Developers
- When
  - "Crash" Staff Meetings
  - TSA Road Show
  - Town Halls
  - Continuous Communications
    - Bagels 'n' Code
    - Lunch 'n' Learn
- How
  - Posters
  - Confluence Site
  - Training Modules
- What
  - Specific Deep Dives
  - Architecture Glossary
  - Data Governance
  - Enterprise Data Governance
  - Application Governance
  - Technical Architecture Governance
    - Project & Structural Governance for each
  - Developer Runbook./Cribsheet

REVIEW DRAFT 2011-01-08A

Attachment C

High Level Roadmap from TSA (1 of 3)

Domain	Name	Project / Project Objective	Visible Benefit	*Key Steps*	2010	2011	2012	2013
Data Lifecycle Management	**01-01 - Establish Data Service Framework**	Develop Data Service Framework to be used by all golden sources (Eliminate SGS)	Elimination of Point to Point Interfaces; Easier integration in WMA Environment	*01-01-01 - Proof of Concept with CPOS (Client Position)*	x			
				01-01-02 - Service Layer redesign	x	x		
	01-02 - Financial Instrument Master	Enhance PRDB in order to enable the elimination of redundant data stores (eliminate SGS)	Single source for financial instrument data	*01-02-01 - Develop requirement-based data architecture and governance model*	x			
	01-03 - Client Master	Enhance CRDB to be WMA's single client master responsible for all client and prospective client reference data (e.g. relationships,etc)	Migrate to client-focussed model away from account centric paradigm.	*01-03-01 - Develop requirement-based data architecture and governance model*	x			
				01-03-02 - Develop execution startegy with various sourcing options	x	x		
				01-03-03 - Implement		x	x	
				01-03-04 - Requirement analysis	x			
				01-03-05 - Data migration		x		
	01-04 - Transaction Master	Develop OET database to be the central orders, executions, and trades data store	Single source for all central orders, executions, and trades that impact an account	*01-04-01 - Develop execution startegy with various sourcing options*	x	x		
	01-05 - Entitlements Master	Consolidate multiple entitlement data stores across the Front Office platform onto a single data store	Enhanced security and entitlement posture; Single control point for entitlement information	*01-05-01 - Implement*		x	x	
	01-06 - Client Transaction, Asset & Liabilities and Transaction Master	Master all client positions, transactions including all assets and liabilities for client data	Full view of client balance sheet and transactions	*01-06-01 - Complete support of liabilities*	x			
				01-06-02 - Develop detailed implementation	x			
				01-06-03 - Implementation (2010: AHA Positions and Liabilities, 2011: AHA Transactions)	x	x	x	
	01-07 - Client Document Master	Single repository for client documents (e.g. Templates, Completed Forms, Communications)	Full view of correspondence with a client	*TBD*		x		
	01-08 - Data Governance	Implement Data Governance process in partnership with the Business owners of the data	Consistent data definitions that can be referenced throughout the organization	*01-08-01 - Model definition and syndication*	x			
				01-08-02 - Process development	x			
				01-08-03 - Model roll-out	x			
	01-09 - Virtual Data Warehouse	Virtual federated data warehouse with common metadata layer, enable enhanced analytics	Enables the consolidation of multiple reporting data stores to enable analytics without the overhead of a physical data warehouse	*01-09-01 - Business Case Validation*		x		
				01-09-02 - Metadata model definition		x		
				01-09-03 - Sourcing option analysis			x	
				01-09-04 - POC			x	
				01-09-05 - Roll-out			x	
	01-10 - Improve Business Intelligence Platform	Ad-hoc data reporting and data mining / analytics (possibly leveraging new data abstraction layer)	Agile data mining, instant analytics	*01-10-01 - Strategy Development*	x			
				01-10-02 - Tool selection and Integration Planning	x			
				01-10-03 - Execution		x		
Application Remediation (Data)	**02-01 - Convert to Universal Account #**	Remediate Applications to Remove business logic from legacy Account number, move onto universal account number	Less prone to data privacy issues (e.g. unautorized access to accounts)	*02-01-01 - Scope definition, planning and costing*		x		
				02-01-02 - Sourcing analysis		x		
				02-01-03 - Remediation roll-out			x	
	02-02 - All Client Data read from CRDB	Remediate applications to read CRDB for client data using the common service	Consistent client data used end to end	*02-02-01 - Application Remediation*	x	x	x	
	02-03 - All Financial Instrument Data read from PRDB	Remediate applications to read PRDB for financial instrument data using the common service	Consistent financial instrument data used end to end	*02-03-01 - Application Remediation*			x	
	02-04 - All Client Transaction Asset & Liability Data read from CPOS	Remediate applications to read CPOS for client position data using the common service	Consistent client position and transaction data presented end to end	*02-04-01 - Application Remediation*		x	x	
	02-05 - All Transaction Data read from	Consildation of orders, executions, and trades						

REVIEW DRAFT 2011-01-08A

Attachment C
High Level Roadmap from TSA (2 of 3)

Post Trade Services	**03-01 - Real-Time Positions and Balances**	Enhance RPB to capture cash balances and consolidate "RPB" like data stores onto single instance (Might be throw away if outsourcing scenario)	Enables Banking Strategy; Real time view of client position and balance	*03-01-01 - Service Definition*	x		
				03-01-02 - Service Implementation and Roll-out		x	x
	03-02 - Remove Margin from Bookkeeping	New Margin Calculation module that isn't embedded with Bookkeeping system and replatform Margin Interest calculation engine (Might be throw away if outsourcing scenario)	Start de-risk bookkeeping platform by de-coupling Margin	*03-02-01 - Requirement and strategy definition*	x		
				03-02-02 - Sourcing and delivery model analysis	x	x	
				03-02-03 - Implementation		x	
				03-02-04 - New Margin system/services deployed		x	x
	03-03 - Replace Core Platform	Replace core functions of the backoffice: Bookkeeping, P&S, Reconciliations, Margin, ACATS, Corporate Actions, Clearing and Settlement	New backoffice platform that can support all products without manual intervention; De-risk backoffice	*03-03-01 - Strategy defition and high level financial model/estimates*	x		
				03-03-02 - Mark to Market / RFI / RFP analysis Go/No Go	x		
				03-03-03 - Execution		x	x
	03-04 - Front to Back Workflow Engine	Deploy a front to back business process workflow infrastructure (in conjunction completing the build out of the Serviceworks platform)	Elimination of Manual Processes	*03-04-01 - Service delivery architecture*	x	x	
	03-05 - Convert to Group Reconciliation Utility	Migrate reconciliations onto TLM with market information	Consistent with Group Tech Startegy; Enable Group Reconciliation Utility	*TBD*	x	x	
	03-06 - Consolidation of Fee Billing Applications	Complete the build out of single fee billing application and decommission legacy applications	Enable different view across client and assest. Provide transparency of revenue structure from accurate billing of fees across products	*TBD*	x	x	
Client & FA Experience	**04-01 - Rewrite Client Frontends**	Re-write OLS under a consistent mobile-enabling framework utilizing roles and entitlements to deliver functionality rather point applications	Consistent look an feel; Improved usability and mobility	*04-01-01 - Market analysis / business requirement*		x	
				04-01-02 - Sourcing option analysis (Including F35 option)		x	
				04-01-03 - Solution development		x	
	04-02 - Rewrite FA Frontends	Re-write Consultworks platforms under a (mobile-enabled) consistent framework utilizing roles and entitlements to deliver functionality rather point applications	Consistent look an feel; Improved usability and mobility	*04-01-01 - Market analysis / business requirement*		x	
				04-01-02 - Sourcing option analysis		x	
				04-01-03 - Solution development		x	
	04-03 - Source all Client Reports from Gold Sources	All front office systems should pull client position and other data from Golden Sources	Consistent view of the client across front-end platforms	*04-02-01 - Model Development (Organizational and Operation) and option analysis*		x	
				04-02-02 - Sourcing option analysis		x	
				04-02-03 - Technology component and process re-engineering			x
	04-04 - Enhanced Account Opening Process	Enhance the account opening process to incoporate Document Management strategy	Enhance client experience	*04-03-01 - Model Development (Organizational and Operation) and option analysis*	x		
				04-03-02 - Sourcing option analysis		x	
				04-03-03 - Technology component and process re-engineering			x
	04-05 - Client Entitlement and Role-Base services	Role based access through a common interfac to provide consistent access and entitlements based upon role		*04-04-01 - Business requirement analysis*	x		
				04-04-02 - Sourcing option analysis		x	
				04-04-03 - Implementation Planning			x
				04-04-04 - Roll-Out			x
	04-06 - Improved Performance Reporting Aligned with Client Goals	Complete PMR enhancements in order to decommission Axys Advent	Integrate performance reporting based on CPOS data	*TBD*	x	x	
	04-07 - Usability, Touch and Feel	Improve the look and feel of Consultworks and OLS	Enhance Client & FA Experience	*04-06-01 - Usability strategy definition*	x		
				04-06-02 - Publishing standards definition and dissimination		x	
				04-06-03 - Remediatioon planning and execution		x	x
	04-08 - Integrated Contact Management Application	Convert remainin FA population onto new CRM system. Integrate CRM application into telephony and Consultworks platform	Industry leading CRM capability	*TBD*	x		
	04-09 - Practice Management Tools	Restructure dashboard and management tool to reflect emphasis on FA benchmarking	Implem,ent new businbess model and strategy with focus on customert goals	*TBD*	x	x	
			"Push to Talk", Integrated				

REVIEW DRAFT 2011-01-08A

Attachment C
High Level Roadmap from TSA (3 of 3)

Order Entry & Trading	05-01 - Consolidate Order Entry Applications	Single Order Entry front end that routes to the appropriate order management application. Consolidate order management applications from 12 to 6	Simplified Order Entry process for the FA; Rationalized application suite	*05-01-01 - Requirement analysis and market survey (product and competition)*		x	
				05-01-02 - Sourcing option analysis (Buy/Build)		x	
				05-01-03 - Implementation		x	x
	05-03 - New Managed Acct Trading Platform	Enhance or replace ICST	True Managed Account Trading Platform that enables true Asset Allocation	*05-03-01 - Scope definition, planning and costing*		x	
				05-03-02 - Sourcing analysis		x	
				05-03-03 - Remediation roll-out		x	x
	05-03 - IB Integration	Leverage IB trading infrastructure (e.g. Oscar, Pinpoint)	Lower cost of ownership; Leverage Group Infrastructure	*TBD*		x	x
	06-01 - One Client Profile	Move towards a wholistic client profile capture methodology and view of assets and liabilities	Full view of a clients portfolio that includes external accounts	*06-01-01 - Opportunity analysis and market survey*	x		
				06-01-02 - Strategy definition		x	
				06-01-03 - Strategy implementation		x	
	06-02 - Integrated Wealth Management Platform	Consolidation of investment tools under an integrated framework that allows for internal and external sourcing of capabilities	Integrate Wealth Planning that enables the FA to ubetter evaluate risks and goals (less product centric)	*06-02-01 - Opportunity analysis and market survey*	x		
				06-02-02 - Strategy definition	x		
				06-02-03 - Strategy implementation		x	x
Banking	07-01 - New Cards Platform	Insource existing credit card platform	Improved Credit Card experience	*TBD*	x	x	
	07-02 - Enhance Mortgage Platform	Upgrade mortage platform to support Business desire to scale the mortgage business	Enable Mortgage Startegy and Integration	*TBD*	x	x	
	07-03 - Upgrade Electronic Billpay	Improve online bill pay experience	Improved Client Experience	*TBD*	x	x	
Products	08-01 - Alternative Investment Automation	Automate AI order entry process to enable straight through processing and scalability	Elimination of manual processes required to support booking of AI investments	*TBD*	x	x	
	08-02 - Enhance Annuities & Insurance Platform	New Insurance and Annuity platform	Able to scale Annuities and Insurance business by eliminating manual process	*TBD*	x	x	
	08-03 - Mutual Fund Back office	New reconciliation tool for Mutual Funds	Enable automated settlement of Mutual Funds	*TBD*	x	x	
	08-04 - Stock Loan		Improve utilization of WMA box	*TBD*	x	x	
	08-05 - Structured Products			*TBD*	x	x	
	08-06 - CEFS	Only upgrade current CEFS platform to meet required commitments to high revenue clients. Focus on consolidation onto WM&SB platform	Reduce cost of mainteing two platforms and compliance with regulatory mandates	*TBD*	x	x	
Supporting Infrastructure Changes	09-01 - OneTransactionKey	Universal transaction ID; consistent immutable internal IDs for any business entity (transactio Traceability client, loan, account)		*09-01-01 - Scope definition, planning and costing*		x	x
				09-01-02 - Sourcing option analysis		x	
				09-01-03 - Rollout / Remediation		x	x
	09-02 - Common Error Handling			*09-02-01 - Scope and effort analysis*		x	
				09-02-02 - Sourcing options analysis		x	
				09-02-03 - Remediation planning and implementation			x

REVIEW DRAFT 2011-01-08A

Attachment D

Staffing Model

Key WMA CTO Process	OWN/SHARE	Cross UBS Impact	Focus	Main Objectives	Total FTE	Management and Senior Project Management
Architecture Review throughout execution of Projects	O		1 - Group CTO	Support Group CTO Quality Process Reviews	0.7	0.1
SDLC Support	S		1 - Group CTO	Support Group CTO Quality Process Reviews	0.4	0.1
				Support Group CTO Quality Process Reviews Total	1.1	0.2
ETC / Category Management WITH WMA	O	X	1 - Group CTO	Promote Group Tech Arch; Category Mgt	0.3	0.1
				Promote Group Tech Arch; Category Mgt Total	0.3	0.1
gCTO Work Groups (Includes Health Care Review)	O		1 - Group CTO	Achieve Group Tech Council Objectives	0.8	0.3
Contribute to CTO Key Processes (Overall)	O	X	1 - Group CTO	Achieve Group Tech Council Objectives	0.4	0.1
Complex Offering Product Management	O		1 - Group CTO	Achieve Group Tech Council Objectives	0.4	0.1
Representing WMA at UBS-level ETC			1 - Group CTO	Achieve Group Tech Council Objectives	0.3	0.2
Strategic Cost Optimization	S	X	1 - Group CTO	Achieve Group Tech Council Objectives	0.2	0.1
				Achieve Group Tech Council Objectives Total	2.1	0.8
Innovation and Other Intellectual Capital Development	O	X	2 - CIO Support	Promote Innovation and R&D	0.7	0.1
				Promote Innovation and R&D Total	0.7	0.1
Reference Architectures Development and Implementation Governance, Syndication	O	X	2 - CIO Support	Drive WMA Target State Architecture	1.2	0.2
Project Management	O		2 - CIO Support	Drive WMA Target State Architecture	1	
Support WMA-Specific Major Initiative	O		2 - CIO Support	Drive WMA Target State Architecture	0.8	0.2
Monitoring implementation of TSA – ie "Air Traffic Control" function	S		2 - CIO Support	Drive WMA Target State Architecture	0.6	0.2
Investment Review / Governance (PRE-TIC)	S		2 - CIO Support	Drive WMA Target State Architecture	0.5	0.1
Maintaining and Promoting TSA	O		2 - CIO Support	Drive WMA Target State Architecture	0.4	0.1
				Drive WMA Target State Architecture Total	4.5	0.8
Reach Out Program to Technologists	S		3 - WMA CTO Fundamentals	Support Vibrant IT Community	0.5	
				Support Vibrant IT Community Total	0.5	0
Promote Group Architectures	O	X	3 - WMA CTO Fundamentals	Support Development of UBS-wide Architectures	0.2	0.1
				Support Development of UBS-wide Architectures Total	0.2	0.1
Suitability Analysis to import technology from/to WMA	O	X	3 - WMA CTO Fundamentals	R&D for new and shared technologies	0.9	0.1
Management of POCs	S		3 - WMA CTO Fundamentals	R&D for new and shared technologies	0.5	
Technology Evaluation	S		3 - WMA CTO Fundamentals	R&D for new and shared technologies	0.4	
Ad hoc research/evaluation of technologies (pre-POC)	O	X	3 - WMA CTO Fundamentals	R&D for new and shared technologies	0.4	
				R&D for new and shared technologies Total	2.2	0.1
Major TSA and non-TSA-related Initiatives	O		3 - WMA CTO Fundamentals	Develop UBS IT Knowledge and Competitive Advantage	0.6	0.1
				Develop UBS IT Knowledge and Competitive Advantage Total	0.6	0.1